Anil Seth

Big Data on a Desktop: A Virtual Machine in an OpenStack Cloud

OpenStack is a worldwide collaboration between developers and cloud computing technologists aimed at developing the cloud computing platform for public and private clouds. Let's install it on our desktop.

Installing OpenStack using Packstack is very simple. After a test installation in a virtual machine, you will find that the basic operations for creating and using virtual machines are now quite simple when using a Web interface.

The environment

It is important to understand the virtual environment. While everything is running on a desktop, the setup consists of multiple logical networks interconnected via virtual routers and switches. You need to make sure that the routes are defined properly because otherwise, you will not be able to access the virtual machines you create.

On the desktop, the virt-manager creates a NAT-based network by default. NAT assures that if your desktop can access the Internet, so can the virtual machine. The Internet access had been used when the OpenStack distribution was installed in the virtual machine.

The Packstack installation process creates a virtual public network for use by the various networks created within the cloud environment. The virtual machine on which OpenStack is installed is the gateway to the physical network.

```
Virtual Network on the Desktop (virbr0 interface):
192.168.122.0/32
IP address of eth0 interface on OpenStack VM: 192.168.122.54
Public Virtual Network created by packstack on OpenStack VM:
172.24.4.224/28
IP address of the br-ex interface OpenStack VM: 172.24.4.225
```

Testing the environment

In the OpenStack VM console, verify the network addresses. In my case, I had to explicitly give an *ip* to the *br-ex* interface, as follows:

```
# ifconfig
# ip addr add 172.24.4.225/28 dev br-ex
```

On the desktop, add a route to the public virtual network on OpenStack VM:

```
# route add -net 172.24.4.224 netmask 255.255.255.240 gw
192.168.122.54
```

Now, browse *http://192.168.122.54/dashboard* and create a new project and a user associated with the project.

1. Sign in as the admin.
2. Under the Identity panel, create a user *(youser)* and a project (Bigdata). Sign out and sign in as *youser* to create and test a cloud VM.
3. Create a private network for the project under *Project/Network/Networks:*
   - Create the private network 192.168.10.0/24 with the gateway 192.168.10.254
   - Create a router and set a gateway to the public network. Add an interface to the private network and *ip* address 192.168.10.254.
4. To be able to sign in using ssh, under the *Project/Compute/Access & Security*, in the *Security Groups* tab, add the following rules to the default security group:
   - Allow ssh access: Custom TCP Rule for allowing traffic on Port 22.
   - Allow icmp access: Custom ICMP Rule with Type and Code value -1.
5. For password-less signing into the VM, under the *Project/Compute/Access & Security,* in the *Key Pairs* tab the following:
   - Select the *Import Key Pair* option and give it a name, e.g., 'desktop user login'.
   - In your desktop terminal window, use ssh-keygen to create a *public/private* key pair in case you don't already have one.
   - Copy the contents of *~/.ssh/id_rsa.pub* from your desktop account and paste them in the public key.
6. Allocate a public IP for accessing the VM under *Project/Compute/Access & Security* in the *Floating Ips* tab, and allocate IP to the project. You may get a value like 172.24.4.229
7. Now launch the instance under *Project/Compute/Instance:*